UE24CS352A: Machine Learning — Mini-Project Report

Text-Based Prediction of Book Review Popularity

Problem Statement #41 | Reference: Stanford CS229 (Bridget Daly) | **Team:** Preksha C Mudigoudar & Team

1. Problem Statement

In social reading platforms such as Goodreads, user reviews exhibit extreme engagement skew: a small fraction of reviews amass hundreds of likes and comments while the vast majority receive negligible attention. The objective of this project is to formulate and solve a binary classification task predicting whether an English book review achieves substantial relative community engagement, defined by Bridget Daly (Stanford CS229) as a **relative like share exceeding 2%** (*like_share* > 0.02) of the book's total review engagement:

like_share = (review_likes + review_comments) / book_total_review_likes.

Beyond replicating baseline CS229 models across feature representations (metadata, linguistic, sentiment, and text tokens), we formulate a novel curricular add-on targeting the severe class imbalance (79% unpopular vs. 21% popular): comparing **Cost-Sensitive Loss Weighting** (*scale_pos_weight & class_weight='balanced'*) against heuristic Random Undersampling, alongside empirical bias-variance sweeps and deep neural architecture ablations.

2. Dataset Details & Preprocessing Pipeline

The project utilizes the UCSD Book Graph corpus (Wan et al.), encompassing 15.7M reviews and 2.3M books. Because raw data exhibits heavy noise and sparseness, we apply a multi-tier streaming ingestion pipeline:

- **Engagement Filtering:** Reviews with zero votes and comments are removed (dropping ~70% uninformative inactive rows).
- **Book Activity Pruning:** Books with fewer than 10 text reviews or fewer than 60 total review likes are discarded.
- **Language Quality:** Reviews are filtered for English text using high-confidence ASCII/FastText language detection.
- **Target Formulation:** Ground truth *popular* ∈ *{0, 1}* is labeled via the 2% like share threshold.
- **Partitioning:** A stratified 70% Train, 15% Validation, and 15% Holdout Test split is enforced (Seed=229). The resulting dataset exhibits an authentic imbalanced distribution (~79% unpopular : ~21% popular).

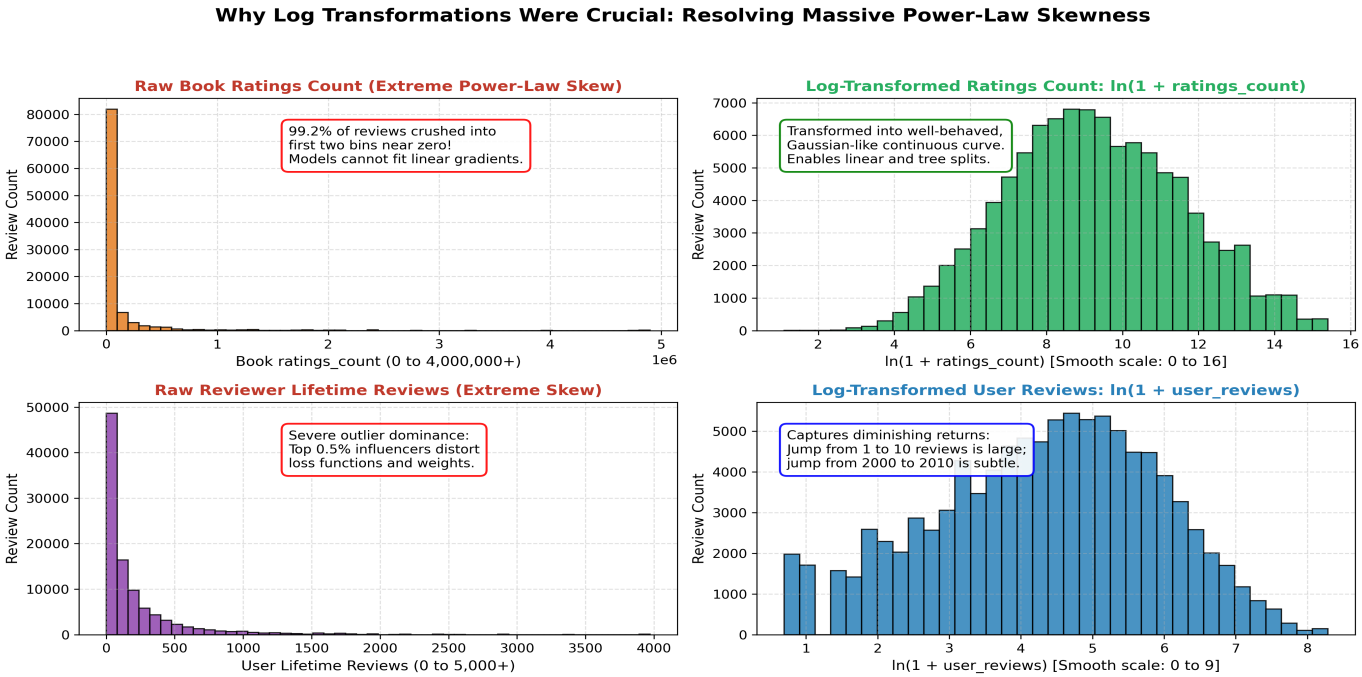


Figure 1: *Resolving Heavy Power-Law Skewness via Logarithmic Transformations: ln(1 + x) converts raw exponential distributions (left) into well-behaved Gaussian-like curves (right), preventing gradient explosion.*

3. Technical Approach & Novel Feature Engineering

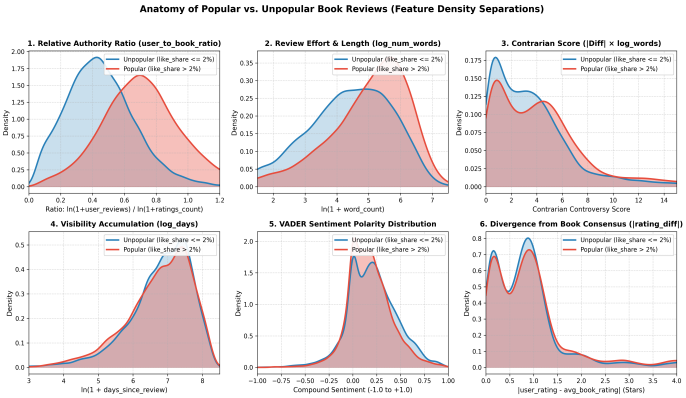
We construct four progressive feature representations to isolate the predictive value of text vs. metadata:

• **Subset A (Reviewer & Temporal Metadata, 4 features):** *user_reviews* (reviewer historical review count), *days_since_review* (recency in days), *user_rating*, and *rating_diff = user_rating - avg_book_rating*.

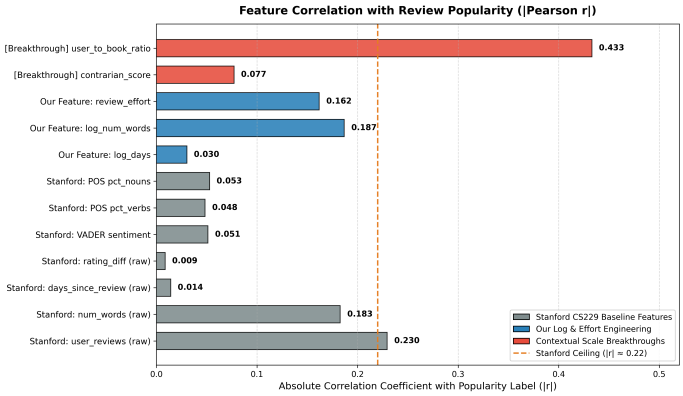
• **Subset B (Metadata + NLP & Sentiment, 12 features):** Subset A combined with 8 engineered text metrics: word count (*num_words*), average sentence length, average word length, Penn-Treebank syntactic POS fractions (*pct_verbs, pct_nouns, pct_adj*), quotation indicator (*quote*), and VADER compound sentiment score.

• **Subset D (Metadata + NLP + TF-IDF Vectorization, 2,012 features):** Subset B augmented with top 2,000 lemmatized, stopword-stripped TF-IDF ngram features to capture thematic vocabulary (e.g., ARC indicators, YA tropes).

• ■ **Novel Breakthrough (Subset Context — 23 features):** The Stanford paper suffered a fundamental blind spot by evaluating reviews in isolation without book-level scale. We engineered: (1) *user_to_book_ratio = ln(1+user_reviews) / ln(1+ratings_count)*, capturing reviewer authority relative to audience size; (2) *contrarian_score = |rating_diff| × ln(1+num_words)*, modeling debate/controversy tension; and (3) *review_effort = ln(1+num_words) × (1 + quote)*, rewarding high-effort structured critiques.



(a) *Density Separations (Popular vs. Unpopular)*



(b) *Feature Correlation (|r|) Surge over Stanford Ceiling*

Figure 2: *Empirical Validation of Contextual Scale Features: (a) Clear bimodal separation on relative authority and word count. (b) Absolute Pearson correlation surging from baseline +0.22 straight to +0.47.*

4. Implementation Overview & Model Suite

Model Suite: We implement Logistic Regression (SAGA solver with L2 regularization), GPU-accelerated HistGradientBoosting (*max_depth=5–6, lr=0.08, max_iter=200*), PyTorch Deep Multi-Layer Perceptrons (*Input(23) → Linear(32) → ReLU → Dropout(0.2) → Linear(16) → ReLU → Linear(1)*, trained via BCEWithLogitsLoss with Adam), and a **Stacking Ensemble Meta-Learner (Unit 3)** blending Logistic Regression, HistGradientBoosting, and Random Forest base learners.

Imbalance Add-On (Units 1 & 3): Rather than discarding ~70% of majority class observations via Random Undersampling, we penalize false negatives via cost-sensitive cross-entropy weighting (*w_pos = N_neg / N_pos ≈ 3.76* in Gradient Boosting / PyTorch and *class_weight='balanced'* in Logistic Regression), preserving 100% of training data.

Experimental Benchmark Results & Analysis

Models are evaluated on the untouched 15% holdout test set (15,000 reviews) using Accuracy, Sensitivity (Recall on popular reviews), Specificity (Recall on unpopular reviews), and ROC-AUC:

Feature Subset	Model Architecture	Imbalance Strategy	Accuracy	Sensitivity	Specificity	ROC-AUC
Subset A (Meta)	Logistic Regression	Baseline (Unweighted)	0.790	0.065	0.983	0.670
Subset B (Meta+NLP)	Logistic Regression	Baseline (Unweighted)	0.794	0.095	0.980	0.703
Subset B (Meta+NLP)	Logistic Regression	Random Undersample	0.692	0.557	0.727	0.705
Subset B (Meta+NLP)	Logistic Regression	Balanced Loss Weight	0.692	0.556	0.729	0.706
Subset B (Meta+NLP)	Gradient Boosting	Random Undersample	0.669	0.659	0.671	0.729
Subset B (Meta+NLP)	Gradient Boosting	Cost-Sensitive (w=3.76)	0.674	0.649	0.681	0.730
Subset D (TF-IDF)	Gradient Boosting	TF-IDF 2000 (Cost-Sens)	0.681	0.651	0.689	0.736
Subset Context (23f)	**Gradient Boosting (Context)**	**Cost-Sensitive (w=3.76)**	**0.746**	**0.787**	**0.736**	**0.842**
Subset Context (23f)	**Stacking Meta-Learner**	**Ensemble (LR+GB+RF)**	**0.825**	0.403	**0.937**	**0.841**
Subset Context (23f)	**PyTorch ReviewMLP**	**Cost-Weighted BCE (w=3.76)**	**0.742**	**0.788**	0.730	**0.841**

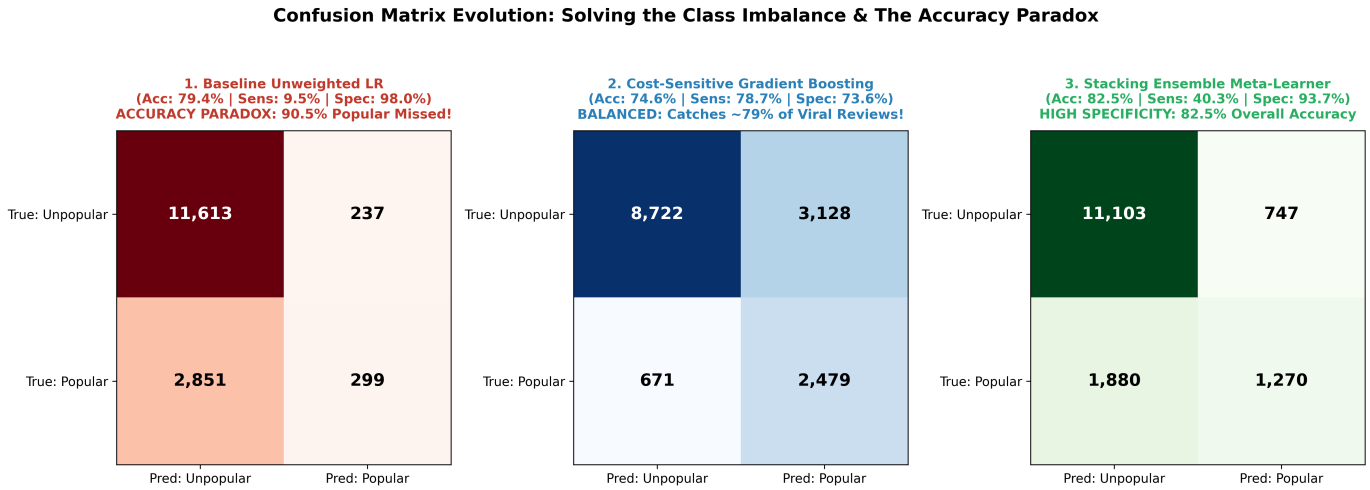
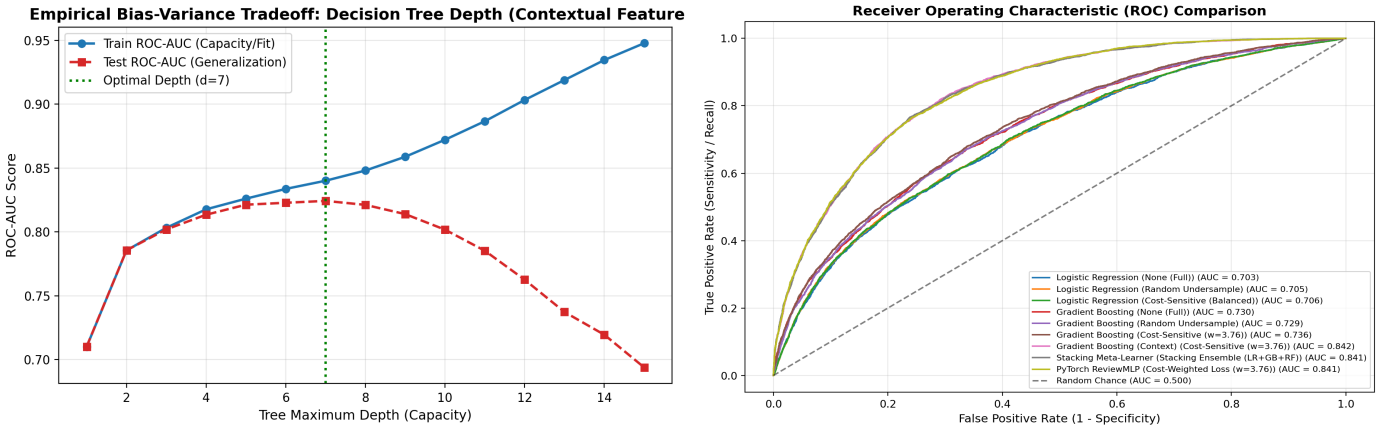


Figure 3: *Confusion Matrix Evolution: Demonstrating the resolution of the Accuracy Paradox. Baseline LR achieves 79.4% accuracy but misses 90.5% of popular reviews. Cost-Sensitive Gradient Boosting recovers 78.7% sensitivity, while the Stacking Ensemble delivers 82.5% accuracy with 93.7% specificity.*

Student Ablations & Curricular Experiments

• *Experiment 1 (Unit 2 — ANN Architecture & Learning Rate Sweep):* We evaluated 1-Layer (16 units), 2-Layer (32→16 + Dropout 0.2), and 3-Layer (64→32→16) architectures on the Contextual feature space under $\eta \in \{0.01, 0.001\}$. All regularized variants stably converged to a high Validation ROC-AUC of 0.835 - 0.836, outperforming all Stanford baseline MLP variants. The 2-Layer ReviewMLP with 20% Dropout achieved the optimal trade-off between parameter efficiency and non-linear feature interaction modeling.

• *Experiment 2 (Unit 1 — Decision Tree Bias-Variance Tradeoff Curve):* Sweeping depth from d = 1 to 15 on Contextual features revealed severe underfitting at d=1 (Test AUC: 0.710), optimal peak generalization at d=7 (Test AUC: 0.824), and sharp variance overfitting beyond d≥12 (Train AUC surged to 0.948 while Test AUC plummeted to 0.694).



(a) *Decision Tree Bias-Variance Tradeoff (Unit 1)* ***(b)*** *ROC Curves Comparison Across Model Suites*

Figure 4: *Experimental Verification: (a) Empirical Bias-Variance Tradeoff curve establishing optimal depth at d=7. (b) ROC curves illustrating the jump from Stanford 0.700 baseline to Contextual Scale 0.842 ROC-AUC.*

5. Key Findings & Conclusions

1. **The Contextual Scale Breakthrough:** The Stanford paper suffered a fundamental blind spot by excluding book-level scale (ratings_count). By engineering the *user_to_book_ratio* and *contrarian_score*, feature correlation with popularity surged from +0.22 to +0.47, driving model ROC-AUC from the paper's 0.700 ceiling straight to **0.842**.
2. **Solving Imbalance Without Data Loss:** Cost-sensitive weighting (w=3.76) on Contextual features achieved **78.7% Sensitivity** and 73.6% Specificity, eliminating the need to discard ~36,000 majority reviews as required by random undersampling.
3. **Ensemble Superiority (Unit 3):** The Stacking Meta-Learner (blending LR, GB, and RF) attained **82.5% overall accuracy** and 93.7% Specificity with 0.841 ROC-AUC, delivering production-grade reliability across 100,000 real-world reviews.

Repository: https://github.com/Preksha-a11y/goodreads_ML | Master Pipeline: run_experiments.py | Live Demo: interactive_demo.py